INCH-POUND

MIL-C-71189 (AR)
AMENDMENT 3
12 April 2001
SUPERSEDING
AMENDMENT 2
31 October 1996

MILITARY SPECIFICATION

CARTRIDGE, 81MM, INFRARED ILLUMINATING, XM816
LOADING, ASSEMBLING AND PACKING

MIL-C-71189 was inactivated after 20 February 1997 for new design.

This amendment forms a part of MIL-C-71189 (AR), dated 23 June 1994, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2, Under Drawings: Add:

"12977137 - Packing and Marking for Container, Metal, M548 for Canister Loading Assembly
for Cartridge, M853A1, 81mm and Canister Loading Assembly for Cartridge, XM816, 81mm".

PAGE 3

* 3.4, Delete in its entirety and substitute the following:

"3.4 Static functioning of the illuminant assembly. The static burning time shall be not less than 70 seconds and no more than 95 seconds. The average infrared output, defined as 0.6 to 0.9 micron wavelength in the near infrared spectrum, shall be not less than 75 watts per steradian for the best for the best 60 seconds after the initial 7 seconds of burning time from ignition and the average visible light output for the same best 60 seconds shall not exceed 500 candela. Determination shall be as specified in 4.4.5.4."

AMSC N/A FSC 1315
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PAGE 4

* 3.6, Delete in its entirety and substitute the following:

"3.6 Pull test for the oval sleeve crimps to suspension wire and suspension wire ring. The oval sleeve crimps to the suspension wire and suspension wire ring shall withstand a pull force of 2,000 pounds minimum."

PAGE 7

Under TABLE II. First article inspection, delete drawing number "dwg. 12953387" and substitute "dwg. 12977252" wherever it appears.

PAGE 11

* 4.4.2.1, Delete "12953387" and substitute "12977252"

PAGE 12

4.4.2.2, Delete drawing number "12953387" and substitute "12977252".

PAGE 13

4.4.2.3, Delete drawing number and substitute "12953387" to "12977252".

PAGE 14

* 4.4.2.4, Major 104, Delete in its entirety and substitute the following:

"Major 104 Pull test for oval sleeve crimps to suspension wire and suspension ring . . . 1..3.6..4.4.5.10"

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* 4.4.2.5:

Delete Special A in its entirety.

Delete Special B in its entirety.

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Delete NOTE 1 in its entirety and substitute the following:

"1/ See 4.4.3.4"

Delete Major 109 in its entirety and substitute the following:

"109 Gap, max between tail cone and body tube...III...3.2...Gage"

Major 112, conformance criteria, delete "Level III".

Add Major 113, 114, 115, 116 as follows:

"113 Warning missing, incomplete, or incorrect... 100%...3.2...Visual
114 Obturator ring missing 1/... 100%...3.2...Visual
115 Gap, max between tail cone and body tube ...III...3.2...Gage
116 Obturator ring pull test 5/... 3.2...Gage/4.4.5.8"

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* 4.4.2.7, Add new Major 109 as follows:

"Major 109 Obturating ring missing...100%...3.2..Visual"

PAGE 20

4.4.2.8, Majors 104, 106 and 107: Delete in their entirety.

PAGE 23

4.4.2.12: Add the following as included page 23a.

4.4.2.13: Add the following as included page 23b.

4.4.2.14: Add the following as included page 23c.

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QUALITY CONFORMANCE INPSECTION
CLASSIFICATION OF CHARACTERISTICS MIL-C-71189 (AR)
AMENDMENT 3

4.4.2.12	TITLE Packing and Marking for Container, Metal, M548 for Canister Loading Assembly for Cartridge, M853A1/XM816 81mm (Prior to Closing)	SHEET 1 OF 1	DRAWING NUMBER 12977137

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMEN PARAGRAPH	NEXT HIGHER ASSEMBLY INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Canister loading assembly improperly packed	Level III	3.2	Manual/Visual
Minor				
201	Packing material in unusable condition	Level V	3.2	Visual
202	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

QUALITY CONFORMANCE INPSECTION
CLASSIFICATION OF CHARACTERISTICS MIL-C-71189 (AR)
AMENDMENT 3

4.4.2.13	TITLE Packing and Marking Container, Metal, M548 for Canister Loading Assembly for Cartridge, M853A1/XM816 81mm (After Closing)	SHEET 1 OF 1	DRAWING NUMBER 12977137

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Contents of metal can loose	Level III	3.2	Manual
102	Cover assembly not secure	Level III	3.2	Manual
Minor				
201	Marking missing, misleading or unidentifiable	Level V	3.2	Visual
202	Metallic seal missing or improperly installed	Level V	3.2	Visual
203	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:

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4.4.2.14	TITLE Support, Propelling Charge	SHEET 1 OF 1		DRAWING NUMBER 12977198
				NEXT HIGHER ASSEMBLY 9362799/
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	12630600 INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Width of opening	1/	3.2	Gage
102	Inside length	1/	3.2	Gage
103	Radius to outside surface of rib (3 places)	1/	3.2	Gage
104	Inside radius to support wall	1/	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.8	Visual

NOTES:
1/ Sample three (3) parts from each mold cavity at the start of each mold run. All dimensions shall be as specified in the drawing prior to production run.

PAGE 24

* 4.4.3.2, Delete in its entirely and substitute the following:

"4.4.3.2, Static functioning. The test shall be performed as specified in 4.4.5.4. A sample for the First Article Inspection is specified in the Table II. A sample for the inspection lot shall be selected in accordance with the sampling plan shown in 4.4.3.2.a. The candle loading assemblies shall be tested for static burning time and candle ouput. If the static burning time of any individual test sample is less than 70 seconds or more than 95 seconds, or the average infrared output for the best 60 seconds of the sample mean is less than 75 watts per steradian, or the average visible output for the same best 60 seconds is more than 500 candela, the inspection lots represented by the samples shall be rejected. No individual test sample shall have the average infrared output for the best 60 seconds less than 70 watts per steradian or the average visible output more than 525 candela for the same best 60 seconds, otherwise the inspection lot shall be rejected."

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* 4.4.3.4, Delete in its entirety and substitute with the following:

"4.4.3.4 Obturating ring pull test. This test shall be performed at the start of a production run and after each 100 cartridges that are welded during that run. The run shall be continuous for the entire time. If the welding run is stopped for any reason, a new test sample must be tested and the test sampling process above shall resume."

* 4.4.3.5, Delete the title " Pull test for oval sleeve crimps to suspension wire" and substitute " Pull test for oval sleeve crimps to suspension wire and suspension wire ring"

PAGE 31

* 4.4.5.4.1:

Delete the third sentence and substitute the following:

"The average effective radiance is to be calculated, using the best 60 seconds of burn time, by dividing the total integral (watt-sec/steradian) over this period by 60."

Delete the last sentence and substitute the following:

"If the individual burning time or average effective radiance does not meet the requirement of paragraph 4.4.3.2, the illuminant inspection lot represented by the sample shall be rejected."

PAGE 32

* 4.4.5.8, Delete in its entirety and substitute with the following:

"4.4.5.8, Obturating ring pull test. Properly install and weld an obturating band onto a simulated 81mm body groove of the correct size and shape (see tail cone drawing) with a relief for a pull test hook. Apply the required load perpendicular (radically and in a direction away from the center axis) to the weld. If the weld fails below the minimum weld strength requirement, replace the band on all the rounds since the last test. If the weld fails above the maximum weld strength requirement, replace the band on all the rounds since the last test."

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PAGE 33

* 4.4.5.10, Delete in its entirety and substitute the following:

"4.4.5.10 Pull test for oval sleeve crimps to suspension wire and suspension wire ring. The pull test sample is an assembly which consists of a suspension wire ring and a parachute cable as delineated in drawing 12953388, detail C. The test sample shall be assembled on a universal testing machine. One end of the pulling device shall be attached to the suspension wire ring. The fixture that comes into contact with the suspension wire ring shall simulate the pin drawing 9381088. The other end of the pulling device is attached to the cable loop where the parachute shroud lines would be connected. The applied load shall be in an axial direction until the required pull force is exerted. The rate of travel of the machine shall be two inches per minutes maximum."

PAGE 34

5.1, 5.2 and 5.3: Delete in their entirety and substitute:

5.1 Unit packing.

5.1.1 Level A. Unit packing for cartridges shall be in accordance with the requirements of drawing 9362799.

5.1.2 Level B. Same as level A.

5.1.3 Level C. Not applicable.

5.2 Packing.

5.2.1 Level A. Cartridges shall be packed in accordance with the requirements of drawing 12944509.

5.2.2 Level B. Same as level A.

5.2.3 Level C. Not applicable.

5.2.4 Packing for interplant shipment of canister loading assembly and IR canister loading assembly.

5.2.4.1 Level A. Not applicable.

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5.2.4.2 Level B. Packing for interplant shipment of canister loading assemblies and IR canister loading assemblies shall be in accordance with drawing 12977137.

5.2.4.3 Level C. Same as level B.

5.3 Marking.

5.3.1 Level A. Packaged cartridges shall be marked in accordance with the requirements of drawing 9362799. Packed cartridges shall be marked in accordance with requirements of drawing 12944509.

5.3.2 Level B. Same as level A.

5.3.3 Level C. Not applicable.

5.3.4 Marking for interplant shipment of canister loading assembly and IR canister loading assembly.

5.3.4.1 Level A. Not applicable.

5.3.4.2 Level B. Marking shall be in accordance with requirements of drawing 12977137.

5.3.4.3 Level C. Same as level B.

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